

Q1. What is the rationalized form of $\frac{1}{\sqrt{3}}$?

Q2.
 $\frac{3}{\sqrt{3}}$

Q2.
 $\frac{3}{\sqrt{3}}$

Q2.
 $\frac{3}{\sqrt{3}}$

Q2.
 $\frac{3}{\sqrt{3}}$

Q2.
 $\frac{3}{\sqrt{3}}$

Astronomers measure the distance to a planet as $5\sqrt{2}$ million kilometers. What is the rationalized form of $\frac{1}{\sqrt{2}}$?

$\frac{1}{\sqrt{2}}$

$\frac{2}{\sqrt{2}}$

$\frac{1}{\sqrt{2}}$

$\frac{1}{\sqrt{2}}$

$\frac{2}{\sqrt{2}}$

Scientists use the formula

$\frac{1}{\sqrt{a}}$ to calculate a planet's orbital speed. If $a = 4$, what is the rationalized form of $\frac{1}{\sqrt{a}}$?

$\frac{1}{\sqrt{4}}$

$\frac{1}{\sqrt{4}}$

$\frac{1}{2}$

$\frac{2}{\sqrt{4}}$

$\frac{1}{\sqrt{4}}$

What is the rationalized form of $\frac{2}{\sqrt{7}}$?

$\frac{2}{\sqrt{7}}$

$\frac{1}{\sqrt{7}}$

$\frac{7}{\sqrt{7}}$

$\frac{1}{\sqrt{7}}$

$\frac{2}{\sqrt{7}}$

A spacecraft's altitude is measured as $x\sqrt{3}$ meters. If $x = 2$, what is the rationalized form of $\frac{1}{\sqrt{3}}$?

$\frac{1}{\sqrt{3}}$

$\frac{1}{\sqrt{3}}$

$\frac{1}{\sqrt{3}}$

$\frac{1}{\sqrt{3}}$

$\frac{1}{\sqrt{3}}$

Foundation (Jalapeno)

Open Ended Questions (FRQ)

Q9. An astronomer measures the distance to a star as $x\sqrt{11}$ light-years. If $x = 3$, rationalize the denominator of $\frac{1}{x\sqrt{11}}$. Show your work and simplify your answer.

Q10. A physicist uses the formula $E = \frac{1}{2}mv^2$ to calculate a particle's energy, where E is the energy and m is the mass. If $m = 25$, rationalize the denominator of $\frac{1}{\sqrt{2}}$ and simplify your answer. Show your work.

Chef's Tips

- Check for accurate rationalization of denominators
- Verify correct simplification of radicals
- Ensure student work is neat and organized